

CSA0910 – Java Programming

ACTIVITY – 02

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1. Store 15 integer values in an ArrayList and remove duplicate elements without using another collection

```
import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        ArrayList<Integer> list = new ArrayList<>();

        System.out.println("Enter 15 Integer Values:");

        for (int i = 0; i < 15; i++) {

            list.add(sc.nextInt());

        }

        for (int i = 0; i < list.size(); i++) {

            for (int j = i + 1; j < list.size(); j++) {

                if (list.get(i).equals(list.get(j))) {

                    list.remove(j);

                    j--;

                }

            }

        }

        System.out.println("ArrayList After Removing Duplicates:");

        System.out.println(list);

        sc.close();

    }

}
```

```
PS D:\Java> & 'C:\Program Files\Android\Android Studio\jbr\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Sameer G\AppData\Roaming\Cod
e\User\workspaceStorage\86335c3275d0db529c25a954546d7a5\redhat.java\jdt_ws\Java_72da483b\bin' 'Main'
1
45
34
23
12
32
42
31
1
2
3
4
5
8
90
ArrayList After Removing Duplicates:
[1, 45, 34, 23, 12, 32, 42, 31, 2, 3, 4, 5, 8, 90]
PS D:\Java>
```

2. Store employee IDs in a LinkedList and display them in both forward and reverse order

```
import java.util.*;

class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        LinkedList<Integer> emp = new LinkedList<>();

        System.out.print("Enter Number of Employees: ");

        int n = sc.nextInt();

        System.out.println("Enter Employee IDs:");

        for (int i = 0; i < n; i++) {

            emp.add(sc.nextInt());

        }

        System.out.println("Forward Order:");

        for (int id : emp) {

            System.out.print(id + " ");

        }

        System.out.println("\nReverse Order:");

        ListIterator<Integer> it = emp.listIterator(emp.size());

        while (it.hasPrevious()) {

            System.out.print(it.previous() + " ");

        }

    }

}
```

```

        sc.close();
    }
}

```

```

Enter Number of Employees: 4
Enter Employee IDs:
123
456
789
056
Forward Order:
123 456 789 56
Reverse Order:
56 789 456 123
=== Code Execution Successful ===

```

3. Store email IDs entered by the user in a HashSet. Display only unique email IDs and count the total unique entries

```

import java.util.*;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        HashSet<String> emails = new HashSet<>();
        System.out.print("Enter Number of Email IDs: ");
        int n = sc.nextInt();
        sc.nextLine();

        System.out.println("Enter Email IDs:");
        for (int i = 0; i < n; i++) {
            emails.add(sc.nextLine());
        }

        System.out.println("Unique Email IDs:");
    }
}

```

```

        for (String email : emails) {
            System.out.println(email);
        }
        System.out.println("Total Unique Entries: " + emails.size());
        sc.close();
    }
}

```

```

Enter Number of Email IDs: 2
Enter Email IDs:
sameerahmedg666@gmail.com
sameerahmedg6@gmail.com
Unique Email IDs:
sameerahmedg6@gmail.com
sameerahmedg666@gmail.com
Total Unique Entries: 2

=== Code Execution Successful ===

```

4. Count the frequency of each word in a paragraph using a HashMap

```

import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a Paragraph:");
        String paragraph = sc.nextLine();
        String[] words = paragraph.toLowerCase().split(" ");
        HashMap<String, Integer> map = new HashMap<>();
        for (String word : words) {
            if (map.containsKey(word)) {
                map.put(word, map.get(word) + 1);
            } else {

```

```

        map.put(word, 1);
    }
}

System.out.println("Word Frequency:");

for (String word : map.keySet()) {
    System.out.println(word + " : " + map.get(word));
}

sc.close();
}
}

```

```

Enter a Paragraph:
eading regularly is one of the best habits to develop for mental growth. First, it
expands your vocabulary and introduces you to new words in context. Second, it
reduces stress by allowing your mind to escape daily worries. Finally, reading
sharpens your memory and improves focus over time. Ultimately, picking up a book
every day is a simple yet powerful way to boost your brainpower.

Word Frequency:
habits : 1
introduces : 1
improves : 1
focus : 1
simple : 1
your : 4
regularly : 1
time. : 1
up : 1
day : 1
you : 1
new : 1
in : 1
is : 2
develop : 1
it : 2
picking : 1
worries. : 1
ultimately, : 1
daily : 1

```